Normalization

The behavior of machine learning algorithms changes when the input distribution changes.

{Covariate shift }

layer weights changes during training steps → Outputs of Activation functions of each layer change → Inputs to the next layer, (With different Input distribution)

Conclusion:

Input distribution of each layer changes with each step .

The basic idea behind Normalization is to:

Limit covariate shift by normalizing the activations of each layer

Normalization: Transforming the inputs to be **Zero mean and unit variance.**

Normalization allows each layer to:

- **learn on a more stable distribution of inputs,**
- **Accelerate the training of the network (don't stick with very small learning rates).**